

Shatrughan Kumar

Software Development Engineer II (SDE-2) | Full Stack Developer | Backend Engineer

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Professional Summary

Software Development Engineer II (SDE-2) with 2.5+ years of experience building scalable backend systems, AI-powered search platforms, and distributed processing solutions. Led a team of 2-3 developers while delivering large-scale educational platforms serving 50,000+ users, processing 100,000+ document workflows, and managing 400+ GB of operational data. Experienced in system design, event-driven architecture, performance optimization, vector search, and cloud-based integrations.

Technical Skills

Languages: JavaScript, TypeScript, Python

Backend: Node.js, NestJS, REST APIs, JWT, Passport, RBAC, Socket.IO

Frontend: React.js, Next.js

Databases: MongoDB, MySQL, Qdrant Vector Database

Caching & Queues: Redis, Bull Queue, Background Jobs, Queue-Based Processing, Event-Driven Architecture

AI & Search: OpenAI APIs, Large Language Models (LLMs), Vector Embeddings, Semantic Search, Retrieval Systems, Elasticsearch

Cloud & Integrations: AWS S3, WhatsApp Business API, Zoho CRM, Cashfree

Core Competencies: System Design, Distributed Systems, Microservices, Scalable Architecture, Performance Optimization, Database Optimization

Work Experience

Software Development Engineer II (SDE-2)	Mar 2024 – Present
Schoollog (L.B. Micro Technologies)	Jaipur, Rajasthan

- Led a team of 2-3 developers in building scalable educational technology platforms serving 50,000+ students and managing 400+ GB of operational data.
- Designed and implemented a distributed document-generation pipeline using NestJS, Redis, Bull Queue, and Puppeteer that processed 100,000+ result and admit-card documents while reducing generation time by 70%.
- Architected a centralized communication platform supporting WhatsApp, Email, SMS, and in-app notifications with dynamic templates, event-driven triggers, queue-based processing, and campaign automation.
- Automated timetable and bell-management workflows through AscEdupage integration, replacing manual operations with centralized scheduling, audit logging, reporting, and operational analytics.
- Led technical design and code review initiatives for a team of 2-3 developers while optimizing backend performance through MongoDB query tuning, aggregation optimization, indexing strategies, and Redis-based caching for high-volume reporting workloads.

AI Engineering Highlights

- Designed and implemented a vector-search infrastructure using OpenAI Embeddings, Qdrant Vector Database, Redis, and MongoDB.
- Indexed and processed 500,000+ examination questions to enable semantic search, duplicate detection, and intelligent content discovery.
- Developed an AI-assisted learning platform capable of generating contextual explanations from question banks, uploaded images, and assessment content.
- Improved student self-learning experience through automated answer explanation and doubt-resolution capabilities.

Education

Full Stack Web Development	01/2023 - 10/2023
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Masai School, Bengaluru

Bachelor of Arts	6/2016 - 06/2019
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Deen Dayal Upadhyaya Gorakhpur University